```
u1@ubuntu-desktop1:~/Desktop/devops-labs/configmap$ kubectl get configmap
NAME               DATA   AGE
app-config         4      2d1h
kube-root-ca.crt   1      14d
nginx-config       1      2d1h
```

```
u1@ubuntu-desktop1:~/Desktop/devops-labs/configmap$ kubectl describe configmap app-config
Name:         app-config
Namespace:    default
Labels:       <none>
Annotations:  <none>

Data
====
APP_COLOR:
----
blue

APP_ENV:
----
production

APP_PORT:
----
8080

MAX_CONN:
----
100


BinaryData
====

Events:  <none>
```

```
u1@ubuntu-desktop1:~/Desktop/devops-labs/configmap$ kubectl get configmap app-config -o yaml
apiVersion: v1
data:
  APP_COLOR: blue
  APP_ENV: production
  APP_PORT: "8080"
  MAX_CONN: "100"
kind: ConfigMap
metadata:
  creationTimestamp: "2026-03-15T08:30:56Z"
  name: app-config
  namespace: default
  resourceVersion: "116399"
  uid: a8d6af35-e393-4ae0-b38d-75eda665f1de
```

```
u1@ubuntu-desktop1:~/Desktop/devops-labs/configmap$ kubectl logs env-pod
\KUBERNETES_SERVICE_PORT=443
KUBERNETES_PORT=tcp://10.96.0.1:443
WEBAPP_NODEPORT_SERVICE_HOST=10.108.114.106
WEBAPP_SERVICE_PORT_80_TCP_ADDR=10.97.228.174
FRONTEND_PORT=tcp://10.104.135.149:80
FRONTEND_SERVICE_PORT=80
HOSTNAME=env-pod
WEBAPP_CLUSTERIP_SERVICE_HOST=10.106.38.153
SHLVL=1
WEB_SERVICE_PORT=tcp://10.110.198.225:80
WEBAPP_SERVICE_PORT_80_TCP_PORT=80
WEB_SERVICE_SERVICE_PORT=80
HOME=/root
BACKEND_SERVICE_HOST=10.108.137.17
WEBAPP_SERVICE_PORT_80_TCP_PROTO=tcp
APP_PORT=8080
WEBAPP_NODEPORT_PORT=tcp://10.108.114.106:80
WEBAPP_NODEPORT_SERVICE_PORT=80
FRONTEND_PORT_80_TCP_ADDR=10.104.135.149
LOGGER_SERVICE_HOST=10.97.66.71
NGINX_LAB_SERVICE_SERVICE_HOST=10.101.94.163
WEBAPP_CLUSTERIP_SERVICE_PORT=80
WEBAPP_CLUSTERIP_PORT=tcp://10.106.38.153:80
WEB_SERVICE_PORT_80_TCP_ADDR=10.110.198.225
FRONTEND_PORT_80_TCP_PORT=80
FRONTEND_PORT_80_TCP_PROTO=tcp
BACKEND_SERVICE_PORT=80
BACKEND_PORT=tcp://10.108.137.17:80
WEBAPP_NODEPORT_PORT_80_TCP_ADDR=10.108.114.106
WEBAPP_SERVICE_PORT_80_TCP=tcp://10.97.228.174:80
WEB_SERVICE_PORT_80_TCP_PORT=80
WEB_SERVICE_PORT_80_TCP_PROTO=tcp
LOGGER_PORT=tcp://10.97.66.71:80
LOGGER_SERVICE_PORT=80
WEBAPP_CLUSTERIP_PORT_80_TCP_ADDR=10.106.38.153
NGINX_LAB_SERVICE_PORT=tcp://10.101.94.163:80
WEBAPP_NODEPORT_PORT_80_TCP_PORT=80
NGINX_LAB_SERVICE_SERVICE_PORT=80
WEBAPP_NODEPORT_PORT_80_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
BACKEND_PORT_80_TCP_ADDR=10.108.137.17
WEBAPP_CLUSTERIP_PORT_80_TCP_PORT=80
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
FRONTEND_PORT_80_TCP=tcp://10.104.135.149:80
WEBAPP_CLUSTERIP_PORT_80_TCP_PROTO=tcp
KUBERNETES_PORT_443_TCP_PORT=443
BACKEND_PORT_80_TCP_PORT=80
LOGGER_PORT_80_TCP_ADDR=10.97.66.71
KUBERNETES_PORT_443_TCP_PROTO=tcp
NGINX_LAB_SERVICE_PORT_80_TCP_ADDR=10.101.94.163
WEB_SERVICE_PORT_80_TCP=tcp://10.110.198.225:80
BACKEND_PORT_80_TCP_PROTO=tcp
MAX_CONN=100
LOGGER_PORT_80_TCP_PORT=80
APP_COLOR=blue
WEBAPP_NODEPORT_PORT_80_TCP=tcp://10.108.114.106:80
NGINX_LAB_SERVICE_PORT_80_TCP_PORT=80
LOGGER_PORT_80_TCP_PROTO=tcp
NGINX_LAB_SERVICE_PORT_80_TCP_PROTO=tcp
WEBAPP_CLUSTERIP_PORT_80_TCP=tcp://10.106.38.153:80
WEBAPP_SERVICE_SERVICE_HOST=10.97.228.174
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
BACKEND_PORT_80_TCP=tcp://10.108.137.17:80
```

```
u1@ubuntu-desktop1:~$ kubectl exec env-pod -- env | grep APP
APP_COLOR=blue
APP_ENV=production
APP_PORT=8080
WEBAPP_CLUSTERIP_SERVICE_HOST=10.106.38.153
WEBAPP_NODEPORT_PORT=tcp://10.108.114.106:80
WEBAPP_SERVICE_PORT_80_TCP_ADDR=10.97.228.174
WEBAPP_CLUSTERIP_PORT_80_TCP=tcp://10.106.38.153:80
WEBAPP_NODEPORT_PORT_80_TCP_ADDR=10.108.114.106
WEBAPP_SERVICE_SERVICE_HOST=10.97.228.174
WEBAPP_SERVICE_SERVICE_PORT=80
WEBAPP_SERVICE_PORT_80_TCP=tcp://10.97.228.174:80
WEBAPP_NODEPORT_PORT_80_TCP_PORT=80
WEBAPP_NODEPORT_PORT_80_TCP=tcp://10.108.114.106:80
WEBAPP_CLUSTERIP_SERVICE_PORT=80
WEBAPP_CLUSTERIP_PORT_80_TCP_PORT=80
WEBAPP_SERVICE_PORT_80_TCP_PROTO=tcp
WEBAPP_NODEPORT_PORT_80_TCP_PROTO=tcp
WEBAPP_CLUSTERIP_PORT_80_TCP_PROTO=tcp
WEBAPP_CLUSTERIP_PORT=tcp://10.106.38.153:80
WEBAPP_CLUSTERIP_PORT_80_TCP_ADDR=10.106.38.153
WEBAPP_NODEPORT_SERVICE_PORT=80
WEBAPP_SERVICE_PORT=tcp://10.97.228.174:80
WEBAPP_SERVICE_PORT_80_TCP_PORT=80
WEBAPP_NODEPORT_SERVICE_HOST=10.108.114.106
```

```
u1@ubuntu-desktop1:~$ kubectl exec  env-pod -- env | grep MAX
MAX_CONN=100
```

```
u1@ubuntu-desktop1:~$ kubectl logs selective-pod
Color=blue
Port=8080
```

```
u1@ubuntu-desktop1:~$ kubectl exec -it selective-pod -- env
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
HOSTNAME=selective-pod
TERM=xterm
COLOR=blue
PORT=8080
LOGGER_SERVICE_PORT=80
LOGGER_PORT_80_TCP_ADDR=10.97.66.71
WEB_SERVICE_PORT_80_TCP_PROTO=tcp
WEBAPP_SERVICE_SERVICE_HOST=10.97.228.174
WEBAPP_CLUSTERIP_SERVICE_HOST=10.106.38.153
KUBERNETES_SERVICE_PORT_HTTPS=443
KUBERNETES_PORT_443_TCP_ADDR=10.96.0.1
WEB_SERVICE_PORT=tcp://10.110.198.225:80
NGINX_LAB_SERVICE_PORT_80_TCP=tcp://10.101.94.163:80
WEB_SERVICE_PORT_80_TCP=tcp://10.110.198.225:80
FRONTEND_SERVICE_HOST=10.104.135.149
WEBAPP_CLUSTERIP_PORT_80_TCP_PROTO=tcp
KUBERNETES_PORT=tcp://10.96.0.1:443
KUBERNETES_PORT_443_TCP=tcp://10.96.0.1:443
WEBAPP_NODEPORT_PORT_80_TCP_PORT=80
WEBAPP_CLUSTERIP_PORT_80_TCP_ADDR=10.106.38.153
LOGGER_PORT_80_TCP_PROTO=tcp
WEB_SERVICE_SERVICE_HOST=10.110.198.225
WEB_SERVICE_PORT_80_TCP_PORT=80
FRONTEND_PORT_80_TCP_PORT=80
WEBAPP_NODEPORT_SERVICE_HOST=10.108.114.106
WEBAPP_NODEPORT_PORT_80_TCP=tcp://10.108.114.106:80
WEBAPP_SERVICE_PORT_80_TCP_PROTO=tcp
LOGGER_SERVICE_HOST=10.97.66.71
BACKEND_SERVICE_HOST=10.108.137.17
BACKEND_PORT_80_TCP_PORT=80
FRONTEND_PORT=tcp://10.104.135.149:80
NGINX_LAB_SERVICE_PORT_80_TCP_PROTO=tcp
WEBAPP_CLUSTERIP_PORT=tcp://10.106.38.153:80
WEB_SERVICE_SERVICE_PORT=80
FRONTEND_PORT_80_TCP=tcp://10.104.135.149:80
WEBAPP_NODEPORT_SERVICE_PORT=80
WEBAPP_SERVICE_PORT_80_TCP=tcp://10.97.228.174:80
NGINX_LAB_SERVICE_SERVICE_HOST=10.101.94.163
NGINX_LAB_SERVICE_SERVICE_PORT=80
KUBERNETES_PORT_443_TCP_PROTO=tcp
FRONTEND_SERVICE_PORT=80
WEBAPP_CLUSTERIP_PORT_80_TCP=tcp://10.106.38.153:80
FRONTEND_PORT_80_TCP_PROTO=tcp
BACKEND_PORT=tcp://10.108.137.17:80
BACKEND_PORT_80_TCP=tcp://10.108.137.17:80
KUBERNETES_SERVICE_HOST=10.96.0.1
KUBERNETES_SERVICE_PORT=443
WEBAPP_SERVICE_PORT=tcp://10.97.228.174:80
NGINX_LAB_SERVICE_PORT_80_TCP_PORT=80
NGINX_LAB_SERVICE_PORT_80_TCP_ADDR=10.101.94.163
BACKEND_SERVICE_PORT=80
BACKEND_PORT_80_TCP_PROTO=tcp
FRONTEND_PORT_80_TCP_ADDR=10.104.135.149
WEBAPP_NODEPORT_PORT_80_TCP_PROTO=tcp
WEBAPP_CLUSTERIP_SERVICE_PORT=80
LOGGER_PORT=tcp://10.97.66.71:80
LOGGER_PORT_80_TCP_PORT=80
```

```
u1@ubuntu-desktop1:~$ kubectl get secret
NAME        TYPE     DATA   AGE
db-secret   Opaque   3      40h
```

```
u1@ubuntu-desktop1:~$ kubectl describe secret db-secret
Name:         db-secret
Namespace:    default
Labels:       <none>
Annotations:  <none>

Type:  Opaque

Data
====
DB_NAME:      5 bytes
DB_PASSWORD:  11 bytes
DB_USER:      5 bytes
```

```
u1@ubuntu-desktop1:~$ kubectl get secret db-secret -o yaml
\apiVersion: v1
data:
  DB_NAME: bXlhcHA=
  DB_PASSWORD: cEBzc3cwcmQxMjM=
  DB_USER: YWRtaW4=
kind: Secret
metadata:
  creationTimestamp: "2026-03-15T18:28:37Z"
  name: db-secret
  namespace: default
  resourceVersion: "125111"
  uid: 474cbc72-6ddb-46e1-a71e-e0b4689a75dc
type: Opaque
```

```
u1@ubuntu-desktop1:~$ kubectl get secret db-secret -o jsonpath='{.data.DB_USER}' | base64 -d
admin
```

```
u1@ubuntu-desktop1:~$ kubectl logs db-pod
User=admin
DB=myapp
```

```
u1@ubuntu-desktop1:~$ kubectl exec  db-pod -- env | grep DB
DB_NAME=myapp
DB_PASSWORD=p@ssw0rd123
DB_USER=admin
```

```
u1@ubuntu-desktop1:~$ kubectl describe pod db-pod
Name:             db-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.49.2
Start Time:       Sun, 15 Mar 2026 20:51:36 +0200
Labels:           <none>
Annotations:      <none>
Status:           Running
IP:               10.244.1.195
IPs:
  IP:  10.244.1.195
Containers:
  db:
    Container ID:  docker://14307c4e8b76a774b75d18df68b89018ee42b37728b2b938929858c486d33587
    Image:         busybox
    Image ID:      docker-pullable://busybox@sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f
    Port:          <none>
    Host Port:     <none>
    Command:
      sh
      -c
      echo User=$DB_USER && echo DB=$DB_NAME && sleep 3600
    State:          Running
      Started:      Tue, 17 Mar 2026 12:48:47 +0200
    Last State:     Terminated
      Reason:       Error
      Exit Code:    137
      Started:      Tue, 17 Mar 2026 12:46:29 +0200
      Finished:     Tue, 17 Mar 2026 12:48:04 +0200
    Ready:          True
    Restart Count:  5
    Environment Variables from:
      db-secret   Secret  Optional: false
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-s76mx (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-s76mx:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    Optional:                false
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason          Age                   From     Message
  ----    ------          ----                  ----     -------
  Normal  Pulling         25h (x4 over 40h)     kubelet  spec.containers{db}: Pulling image "busybox"
  Normal  Pulled          25h                   kubelet  spec.containers{db}: Successfully pulled image "busybox" in 1.83s (1.83s including waiting). Image size: 4425286 bytes.
  Normal  Created         25h (x4 over 40h)     kubelet  spec.containers{db}: Created container: db
  Normal  Started         25h (x4 over 40h)     kubelet  spec.containers{db}: Started container db
  Normal  Pulled          6m7s                  kubelet  spec.containers{db}: Successfully pulled image "busybox" in 1.806s (5m52.425s including waiting). Image size: 4425286 bytes.
  Normal  Killing         4m31s                 kubelet  spec.containers{db}: Stopping container db
  Normal  SandboxChanged  4m30s (x2 over 40m)   kubelet  Pod sandbox changed, it will be killed and re-created.
  Normal  Pulling         4m25s (x2 over 40m)   kubelet  spec.containers{db}: Pulling image "busybox"
  Normal  Created         3m49s (x2 over 6m7s)  kubelet  spec.containers{db}: Created container: db
  Normal  Started         3m49s (x2 over 6m7s)  kubelet  spec.containers{db}: Started container db
  Normal  Pulled          3m49s                 kubelet  spec.containers{db}: Successfully pulled image "busybox" in 1.804s (35.885s including waiting). Image size: 4425286 bytes.
```